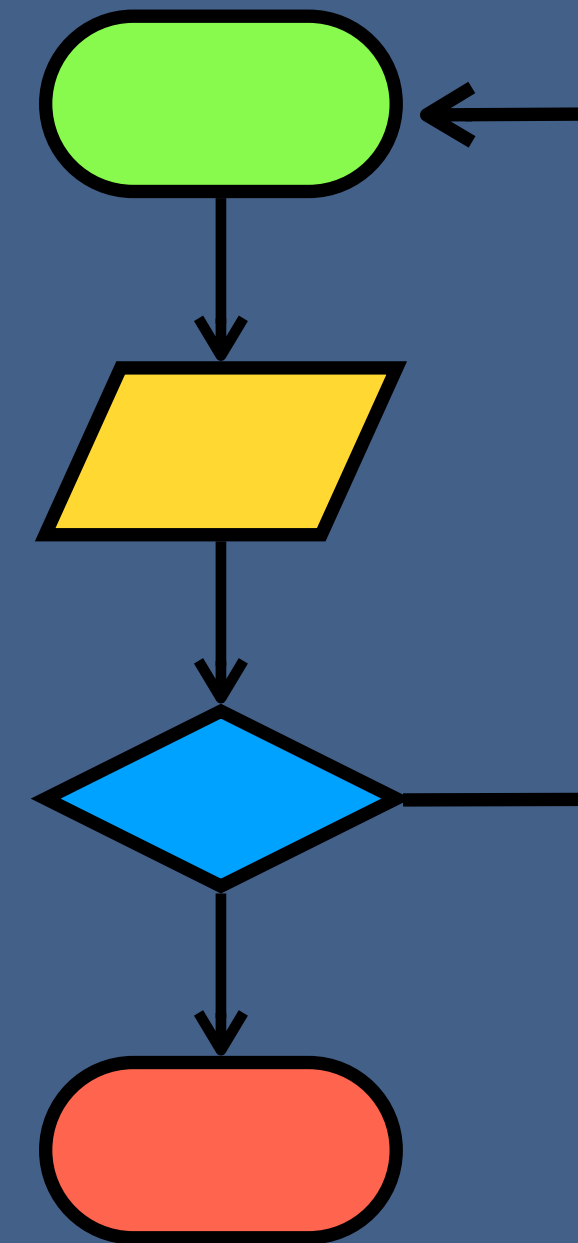
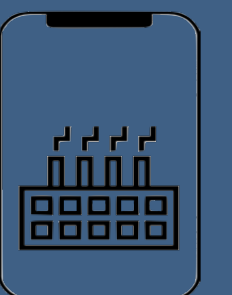
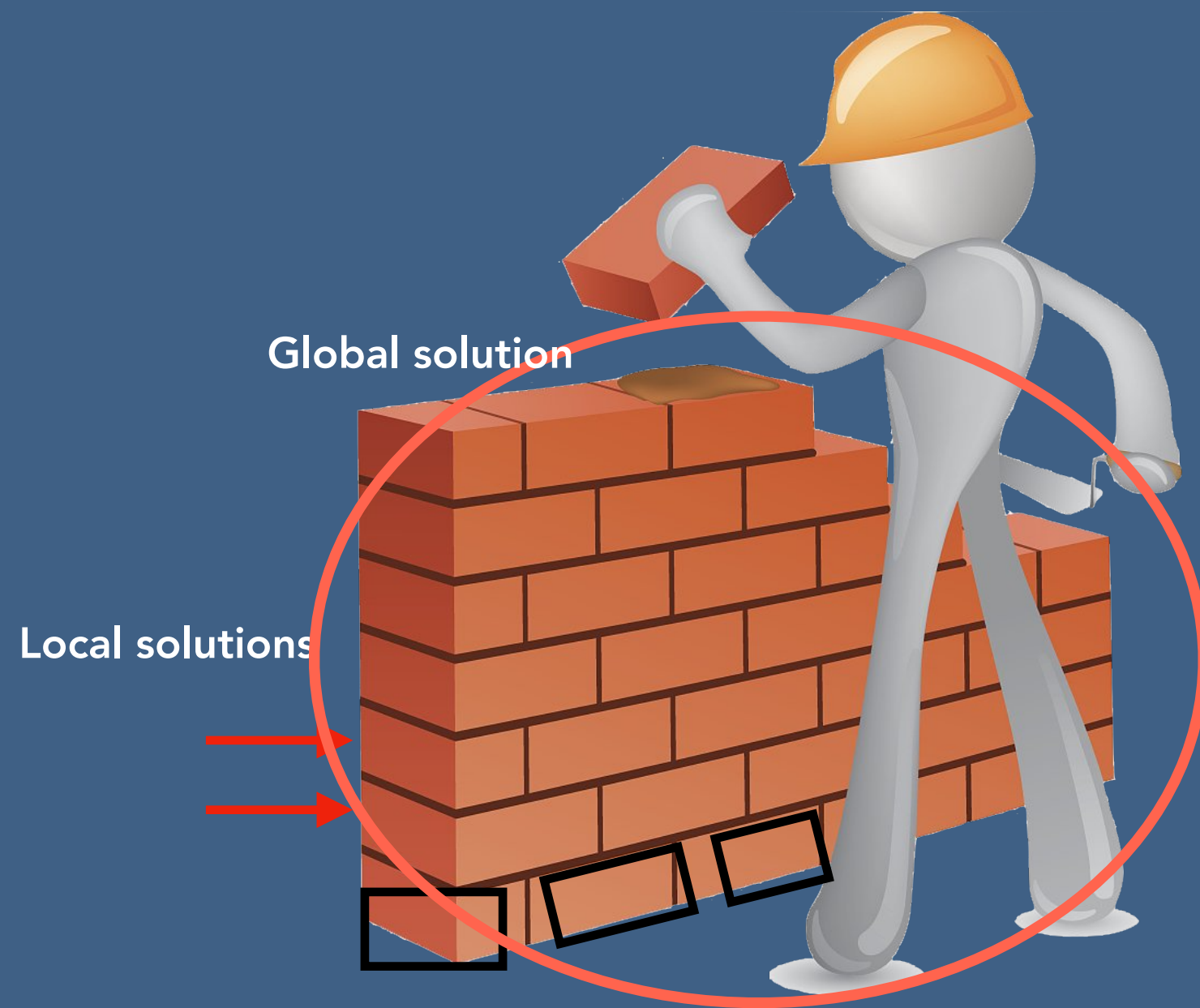


Greedy Algorithms



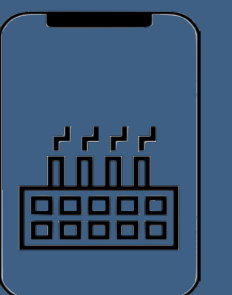
What is Greedy Algorithm?

- It is an algorithmic paradigm that builds the solution piece by piece
- In each step it chooses the piece that offers most obvious and immediate benefit
- It fits perfectly for those solutions in which local optimal solutions lead to global solution



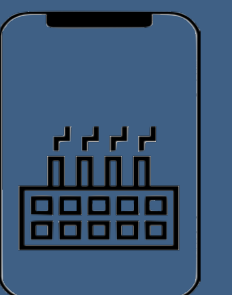
What is Greedy Algorithm?

- Insertion Sort
 - Selection Sort
 - Topological Sort
 - Prim's Algorithm
 - Kruskal Algorithm
-
- Activity Selection Problem
 - Coin change Problem
 - Fractional Knapsack Problem



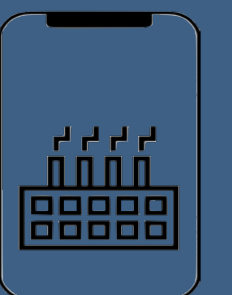
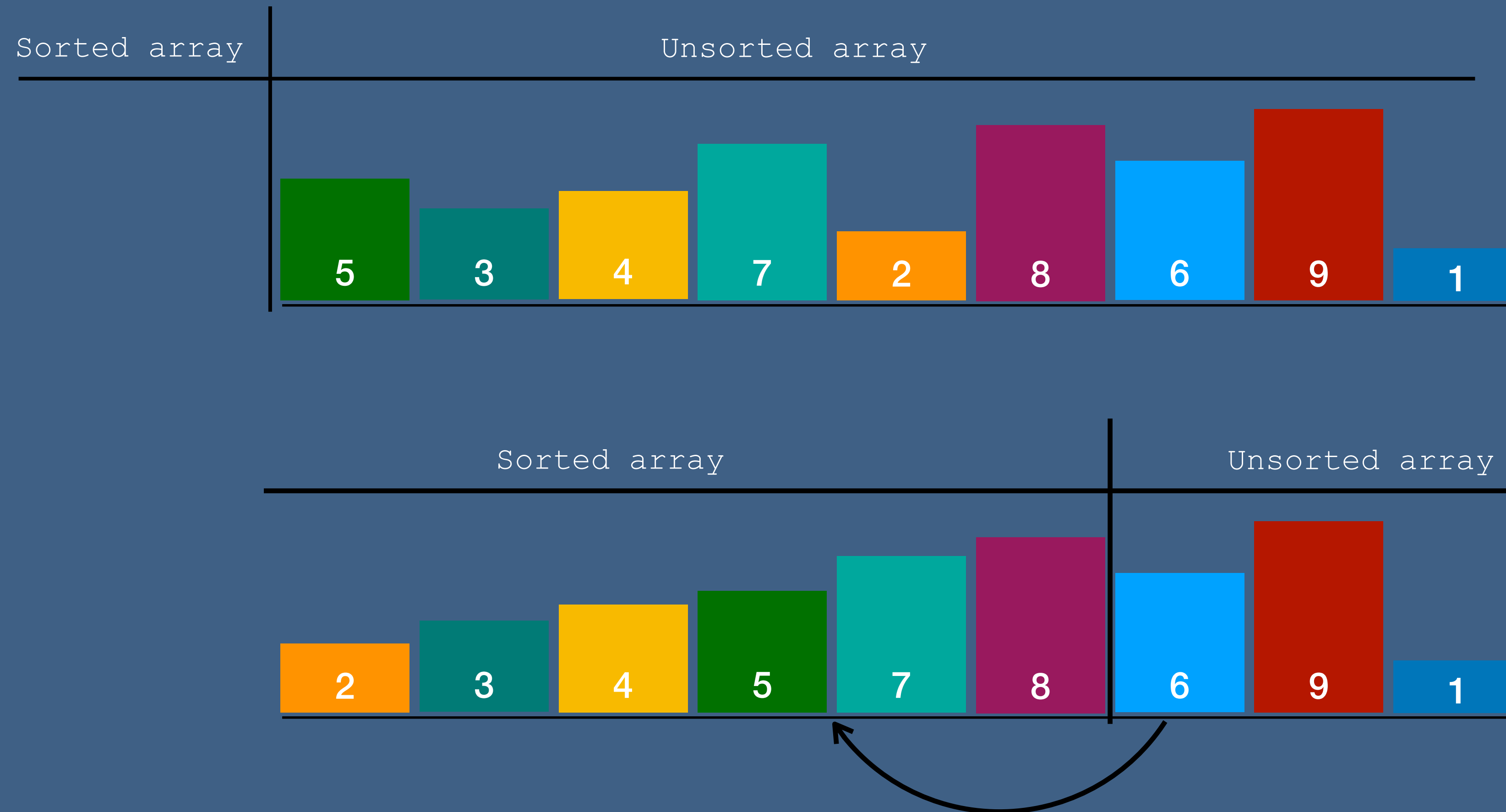
Greedy Algorithms

- Insertion Sort
- Selection Sort
- Topological Sort
- Prim's Algorithm
- Kruskal Algorithm



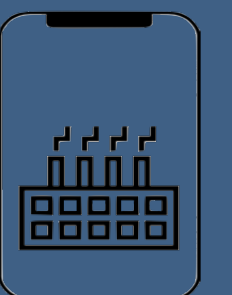
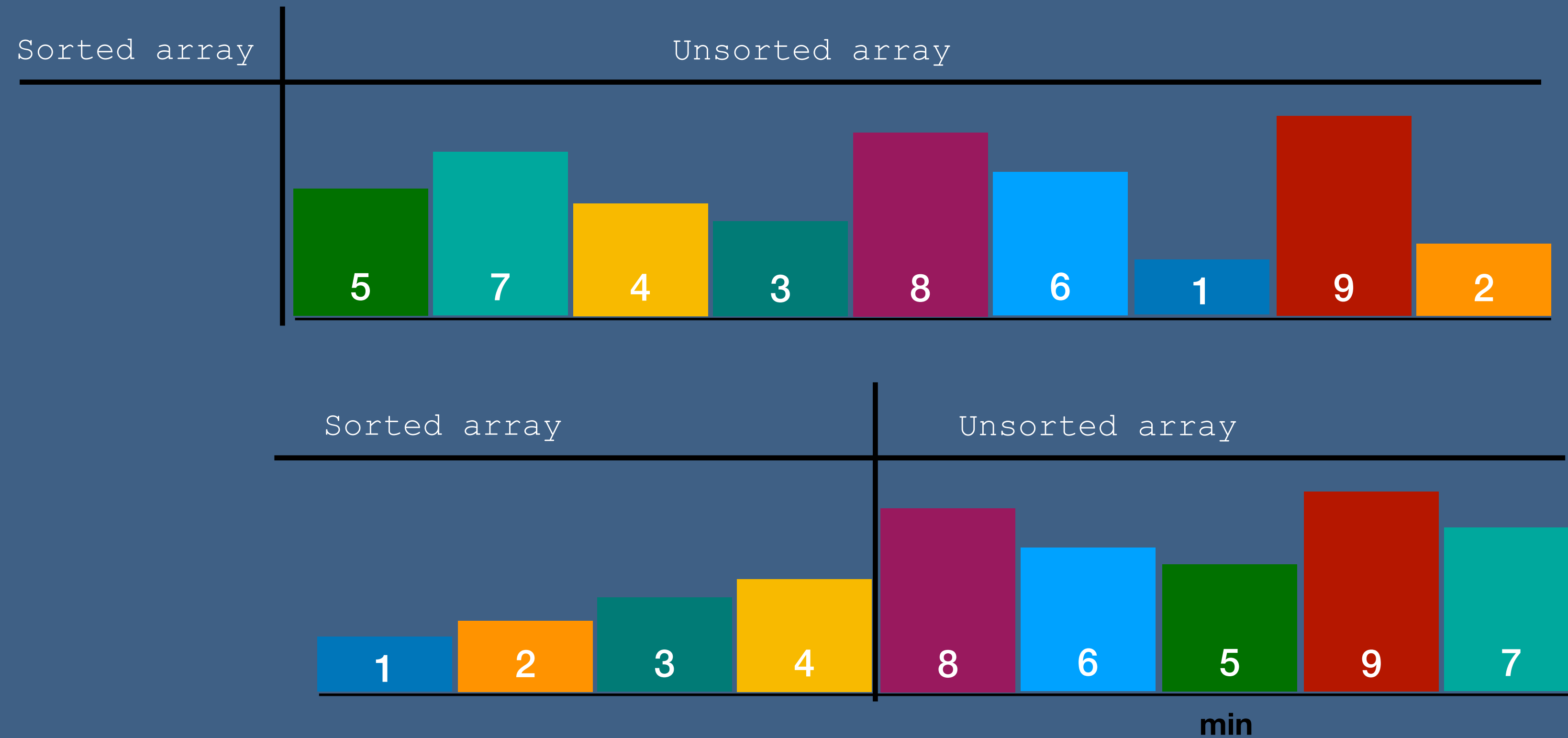
Greedy Algorithms

Insertion Sort



Greedy Algorithms

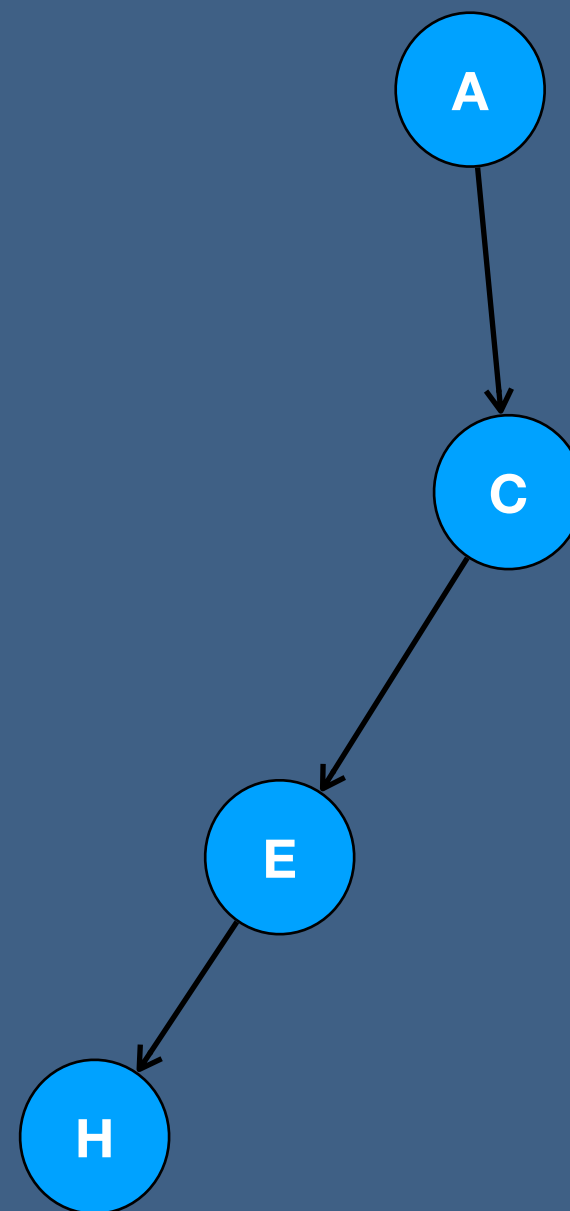
Selection Sort



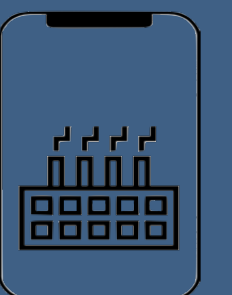
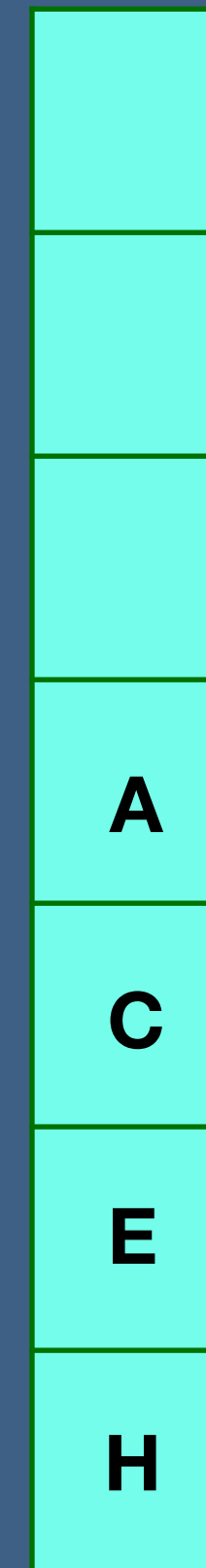
Greedy Algorithms

Topological Sort

A C E H



Stack



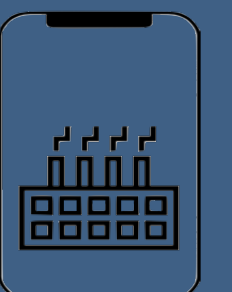
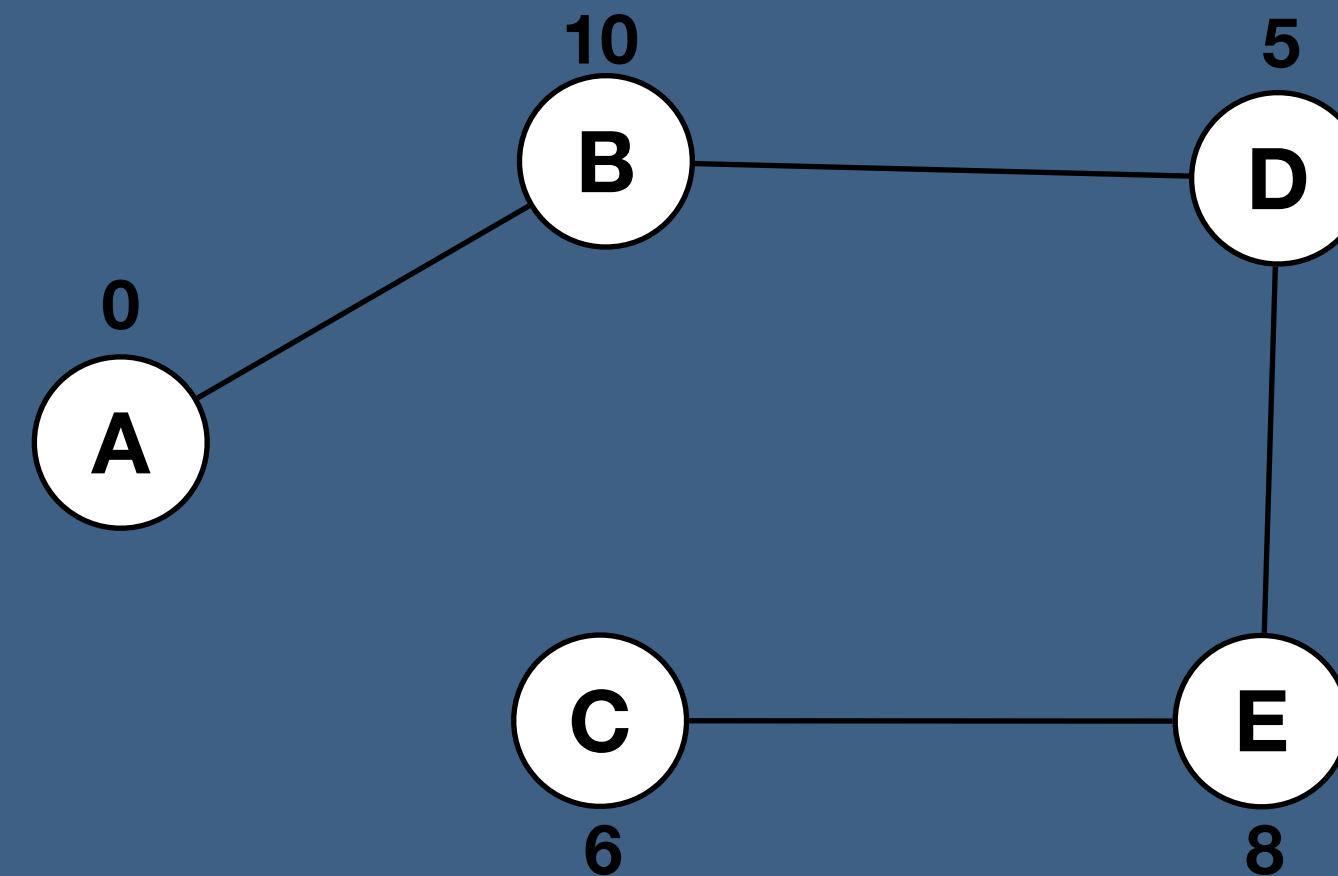
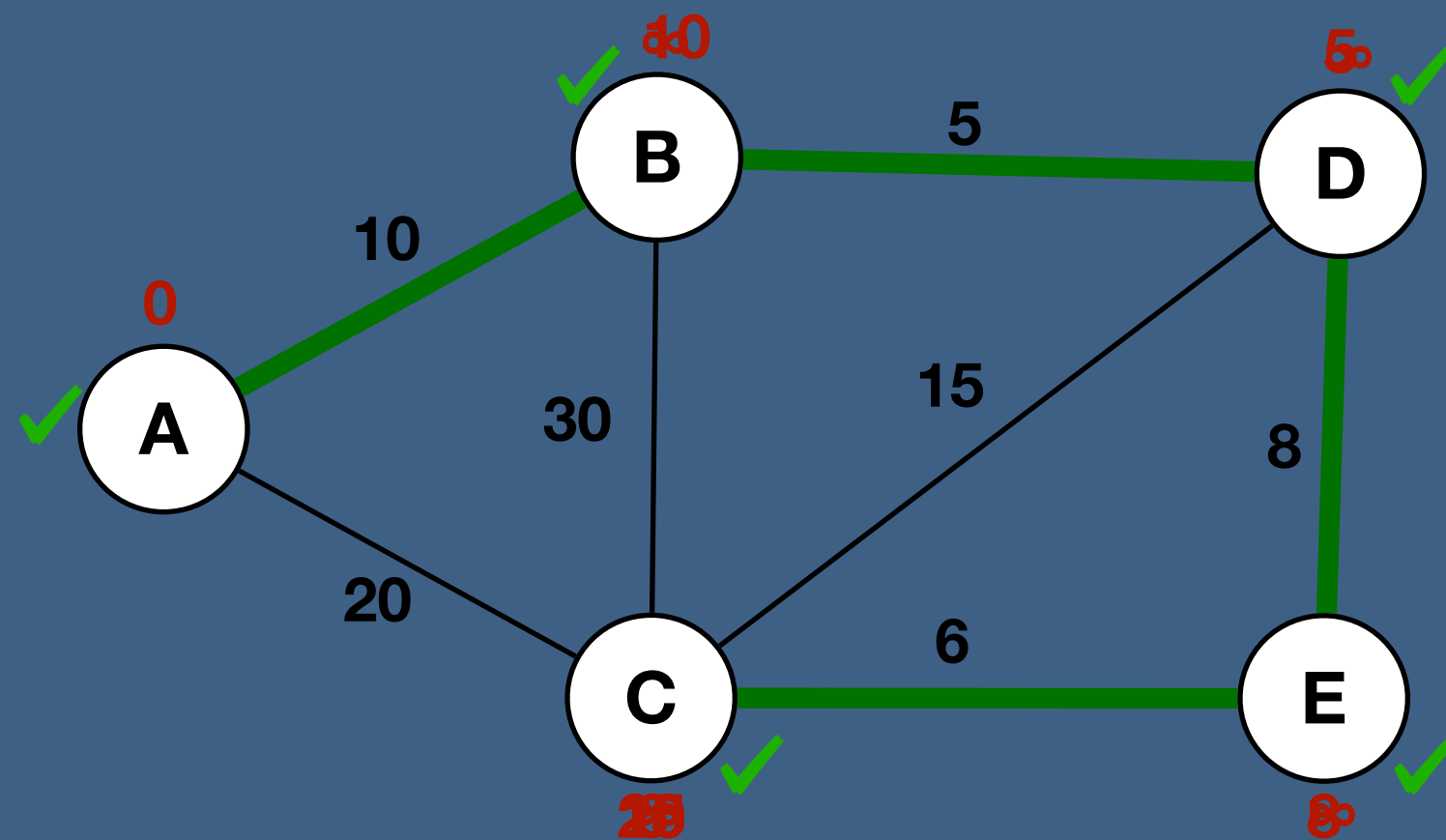
Greedy Algorithms

Prims Algorithm

It is a greedy algorithm

It finds a minimum spanning tree for weighted undirected graphs in following ways

1. Take any vertex as a source set its weight to 0 and all other vertices' weight to infinity
2. For every adjacent vertices if the current weight is more than current edge then we set it to current edge
3. Then we mark current vertex as visited
4. Repeat these steps for all vertices in increasing order of weight



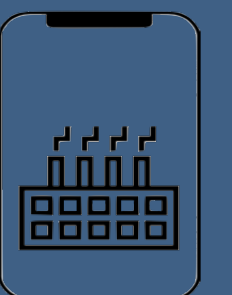
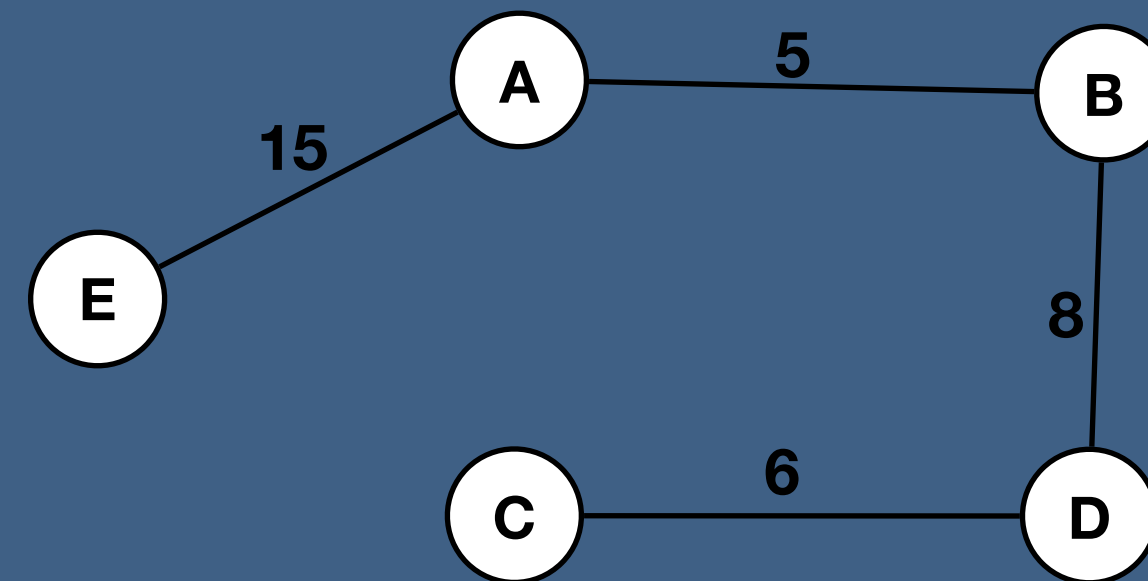
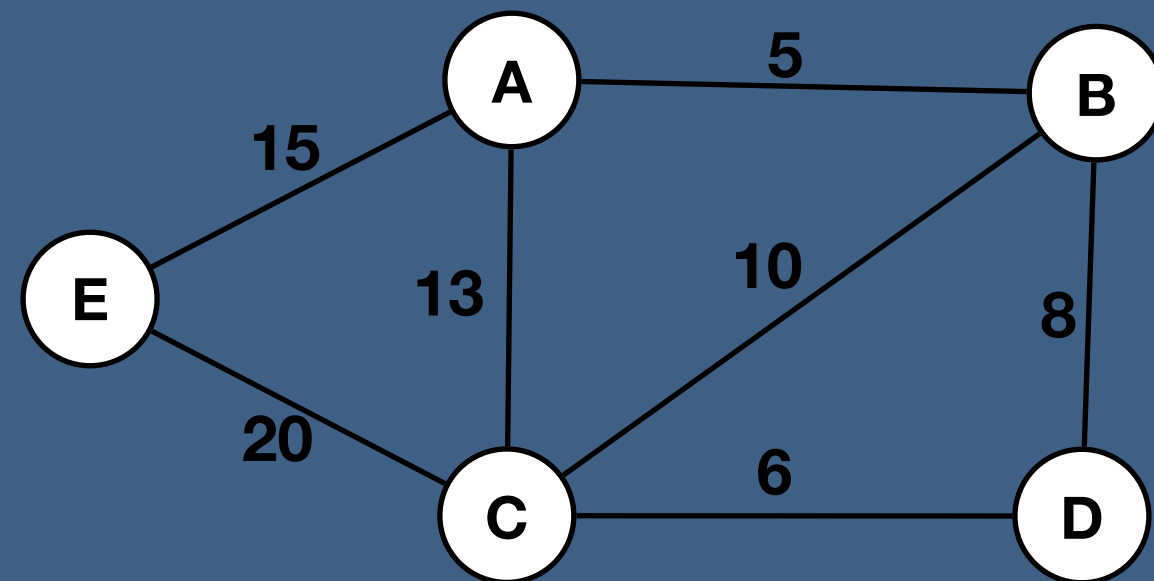
Greedy Algorithms

Kruskal's Algorithm

It is a greedy algorithm

It finds a minimum spanning tree for weighted undirected graphs in two ways

- Add increasing cost edges at each step
- Avoid any cycle at each step



Activity Selection Problem

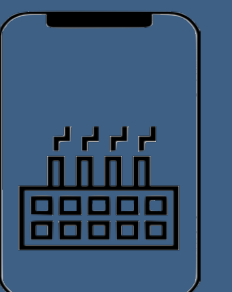
Given N number of activities with their start and end times. We need to select the maximum number of activities that can be performed by a single person, assuming that a person can only work on a single activity at a time.

Activity	A1	A2	A3	A4	A5	A6
Start	0	3	1	5	5	8
Finish	6	4	2	8	7	9

2 Activities

Activity	A3	A2	A1	A5	A4	A6
Start	1	3	0	5	5	8
Finish	2	4	6	7	8	9

4 Activities



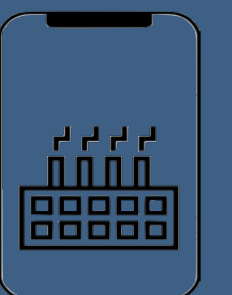
Activity selection problem

Sort activities based on finish time

Select first activity from sorted array and print it

For all remaining activities:

If the start time of this activity is greater or equal to the finish time of previously selected activity then select this activity and print it



Coin Change Problem

You are given coins of different denominations and total amount of money. Find the minimum number of coins that you need to make up the given amount.

Infinite supply of denominations : {1,2,5,10,20,50,100,1000}

Example 1

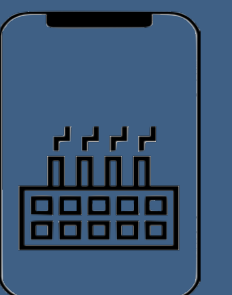
Total amount : 70

Answer: 2 $\rightarrow 50 + 20 = 70$

Example 2

Total amount : 122

Answer: 3 $\rightarrow 100 + 20 + 2 = 122$



Coin Change Problem

You are given coins of different denominations and total amount of money. Find the minimum number of coins that you need to make up the given amount.

Infinite supply of denominations : {1,2,5,10,20,50,100,1000}

Total amount : 2035

$$2035 - 1000 = 1035$$

$$1035 - 1000 = 35$$

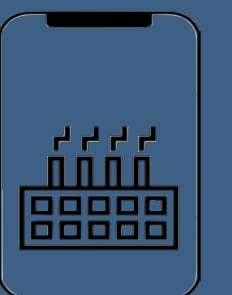
$$35 - 20 = 15$$

$$15 - 10 = 5$$

$$5 - 5 = 0$$

Result : 1000 , 1000 , 20 , 10 , 5

Answer: 5



Coin Change Problem

Find the biggest coin that is less than given total number

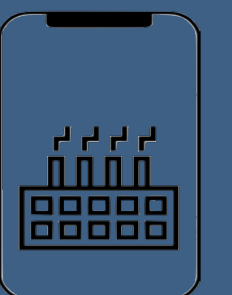
Add coin to the result and subtract coin from total number

If V is equal to zero:

Then print result

else:

Repeat Step 2 and 3



Fractional Knapsack Problem

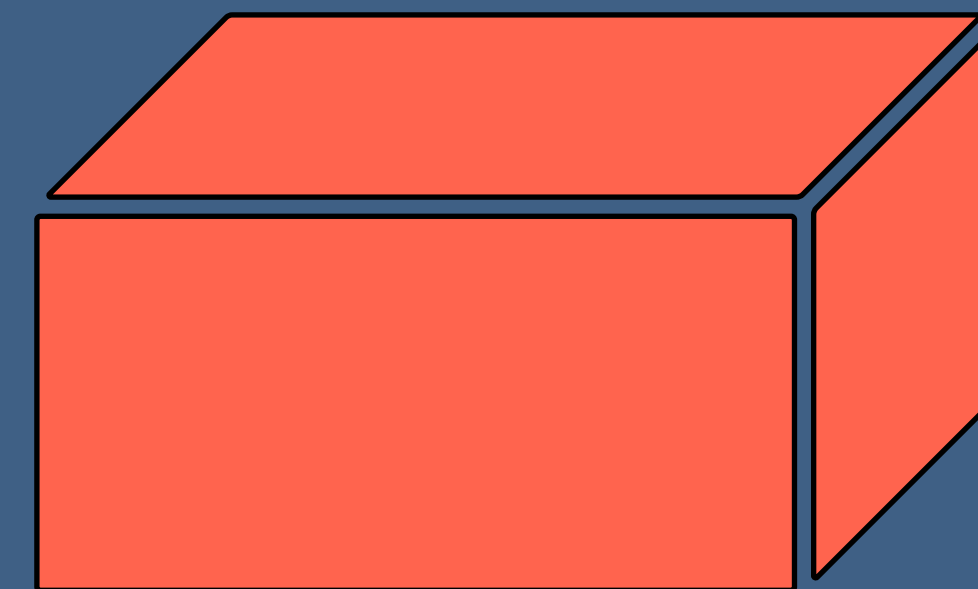
Given a set of items, each with a weight and a value, determine the number of each item to include in a collection so that the total weight is less than or equal to a given limit and the total value is as large as possible.



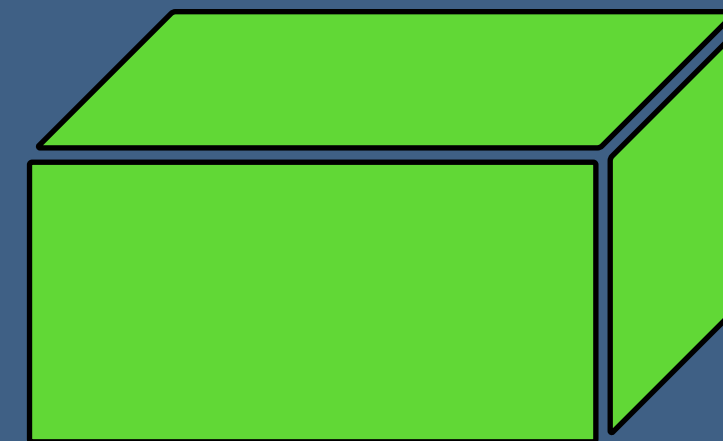
30 kg



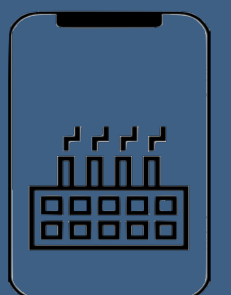
10 kg, Value : 20



20 kg, Value : 10

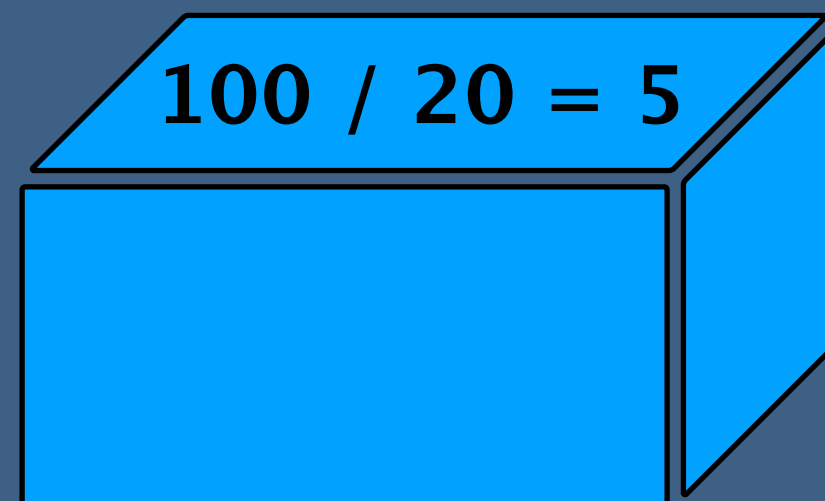


10 kg, Value : 30

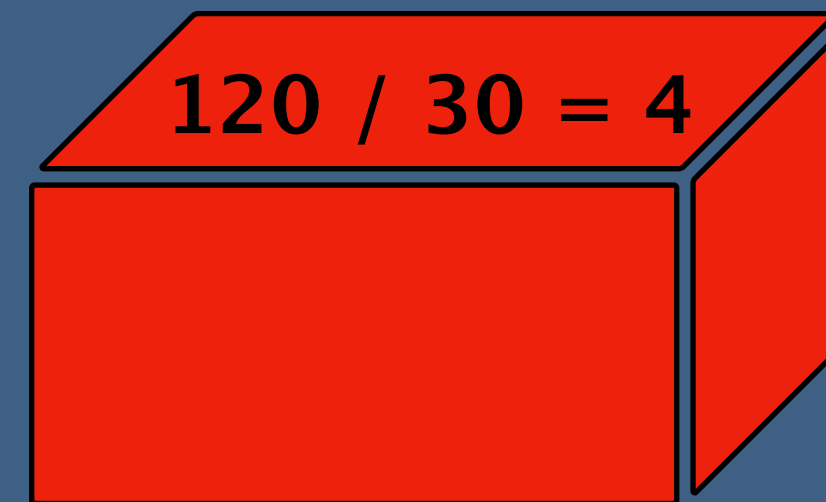


Fractional Knapsack Problem

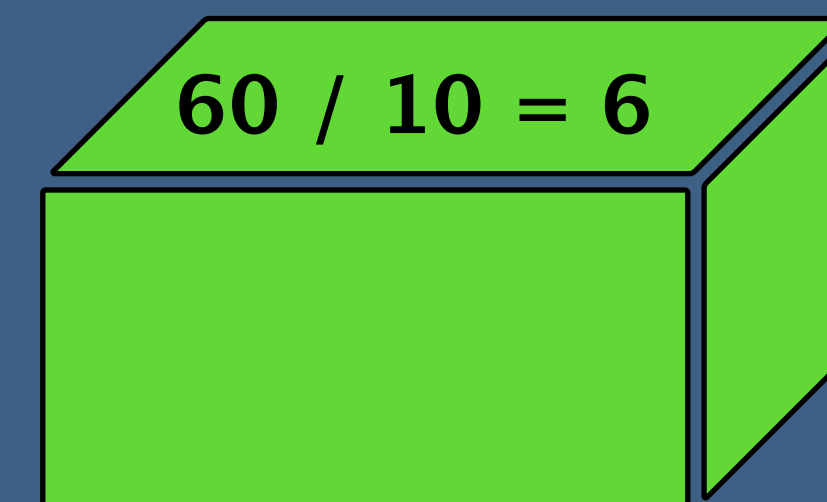
Given a set of items, each with a weight and a value, determine the number of each item to include in a collection so that the total weight is less than or equal to a given limit and the total value is as large as possible.



20 kg, Value : 100



30 kg, Value : 120



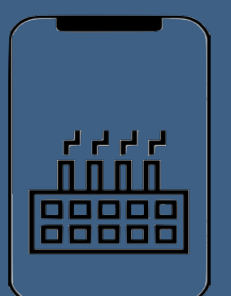
10 kg, Value : 60



50 kg

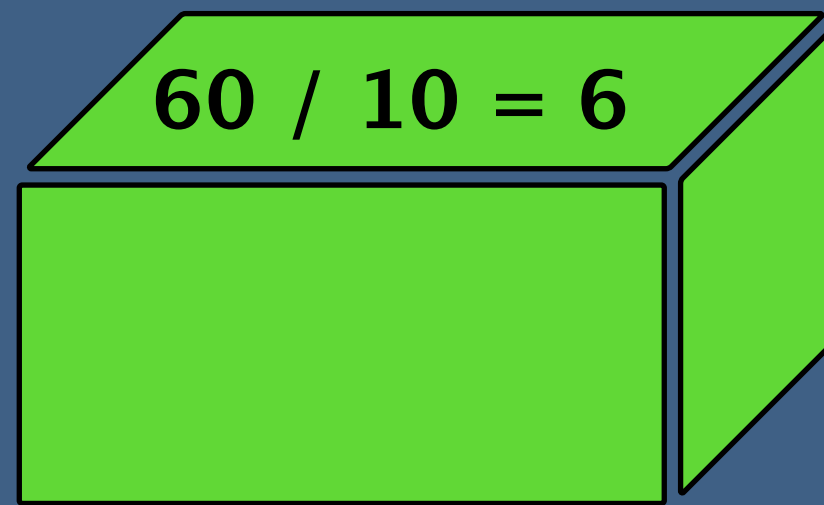
$$100 + 120 = 220$$

$$60 + 100 + 120 * 2/3 = 240$$

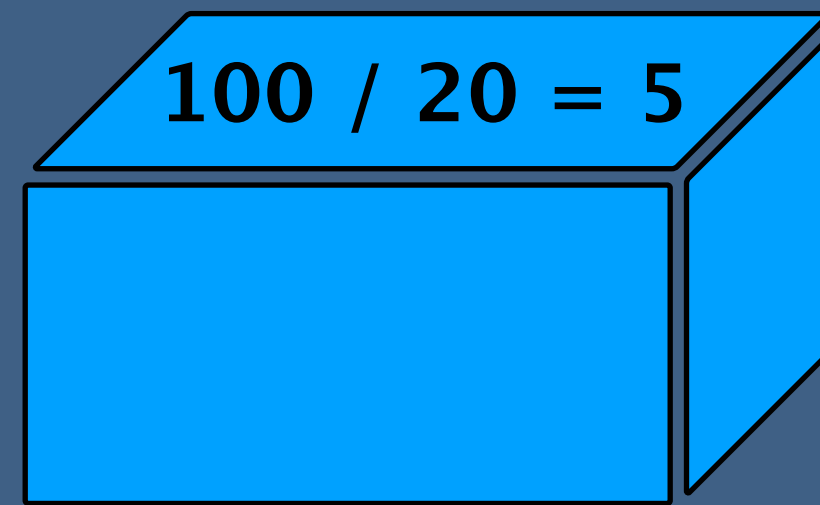


Fractional Knapsack Problem

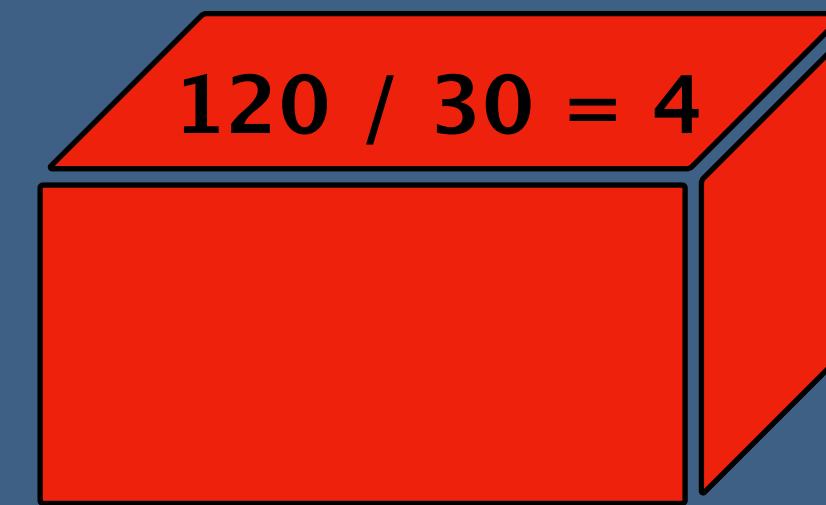
Given a set of items, each with a weight and a value, determine the number of each item to include in a collection so that the total weight is less than or equal to a given limit and the total value is as large as possible.



10 kg, Value : 60



20 kg, Value : 100



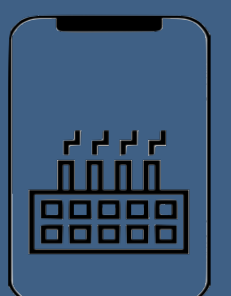
30 kg, Value : 120



50 kg

$$100 + 120 = 220$$

$$60 + 100 + 120 \cdot \frac{2}{3} = 240$$



Fractional Knapsack Problem

Calculate the density or ratio for each item

Sort items based on this ratio

Take items with the highest ratio sequentially until weight allows

Add the next item as much (fractional) as we can

